

WEEK – 3

ER DIAGRAM DESIGN REPORT

Project Title

ShopNest – E-Commerce Order
Management Database System

Submitted By,

S.Yuvasri

II-BCA-“B”

Table of Contents

<i>S.No</i>	Title
1	Introduction
2	ER Diagram Design
3	Entities and Attributes
4	Relationship Design
5	Cardinality and Participation
6	Relationship Mapping
7	Summary
8	Conclusion

1. Introduction

The **ShopNest – E-Commerce Order Management Database System** is designed to manage the core operations of an online shopping platform using a centralized relational database. After completing the Entity and Relationship Analysis in Week 2, the next phase is to design the **Entity Relationship (ER) Diagram**.

The ER Diagram provides a graphical representation of the entities, their attributes, and the relationships among them. It helps in understanding how data is organized and serves as the foundation for converting the conceptual design into a relational schema.

2. ER Diagram Design

The ER Diagram represents all entities identified during the requirement analysis and entity analysis phases. Each entity is connected through appropriate relationships using **Primary Keys (PK)** and **Foreign Keys (FK)**.

The ER Diagram includes the following components:

- CUSTOMER
- CATEGORY
- PRODUCT
- SUPPLIER
- ORDER
- ORDER_DETAILS
- PAYMENT
- SHIPMENT
- REVIEW

The diagram also represents the relationships, attributes, primary keys, foreign keys, and cardinalities required for implementing the database.

3. Entities and Attributes

Entity	Major Attributes
CUSTOMER	Customer_ID, Name, Email, Mobile_Number, Address, Registration_Date
CATEGORY	Category_ID, Category_Name, Description

PRODUCT	Product_ID, Product_Name, Price, Stock_Quantity, Category_ID, Supplier_ID
SUPPLIER	Supplier_ID, Supplier_Name, Contact_Information
ORDER	Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status
ORDER_DETAILS	Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price
PAYMENT	Payment_ID, Order_ID, Payment_Method, Payment_Date, Payment_Status
SHIPMENT	Shipment_ID, Order_ID, Delivery_Address, Shipment_Date, Delivery_Status
REVIEW	Review_ID, Customer_ID, Product_ID, Rating, Comments

4. Relationship Design

The relationships between the entities are defined as follows:

CUSTOMER → ORDER

A customer can place multiple orders, while each order belongs to only one customer.

Relationship: **Places**

CATEGORY → PRODUCT

Each category contains multiple products, while every product belongs to one category.

Relationship: **Contains**

SUPPLIER → PRODUCT

One supplier can supply multiple products, while each product is supplied by one supplier.

Relationship: **Supplies**

ORDER → ORDER_DETAILS

One order contains multiple order details.

Relationship: **Contains**

PRODUCT → ORDER_DETAILS

One product can appear in multiple order details.

Relationship: **Appears In**

ORDER → PAYMENT

Each order has one payment record.

Relationship: **Has**

ORDER → SHIPMENT

Each order has one shipment record.

Relationship: **Has**

CUSTOMER → REVIEW

A customer can write multiple reviews.

Relationship: **Writes**

PRODUCT → REVIEW

A product can receive multiple reviews.

Relationship: **Receives**

5. Cardinality and Participation

Relationship	Cardinality
CUSTOMER → ORDER	One-to-Many (1:M)
CATEGORY → PRODUCT	One-to-Many (1:M)
SUPPLIER → PRODUCT	One-to-Many (1:M)
ORDER → ORDER_DETAILS	One-to-Many (1:M)
PRODUCT → ORDER_DETAILS	One-to-Many (1:M)
ORDER → PAYMENT	One-to-One (1:1)
ORDER → SHIPMENT	One-to-One (1:1)
CUSTOMER → REVIEW	One-to-Many (1:M)
PRODUCT → REVIEW	One-to-Many (1:M)

Participation Constraints

- Every ORDER must belong to one CUSTOMER.
- Every PRODUCT belongs to one CATEGORY.
- Every PRODUCT is supplied by one SUPPLIER.
- Every ORDER contains one or more ORDER_DETAILS.
- Every PAYMENT belongs to one ORDER.
- Every SHIPMENT belongs to one ORDER.
- Every REVIEW belongs to one CUSTOMER and one PRODUCT.

6. Relationship Mapping

Parent Entity	Child Entity	Foreign Key
CUSTOMER	ORDER	CUSTOMER_ID

CATEGORY	PRODUCT	CATEGORY_ID
SUPPLIER	PRODUCT	SUPPLIER_ID
ORDER	ORDER_DETAILS	ORDER_ID
PRODUCT	ORDER_DETAILS	PRODUCT_ID
ORDER	PAYMENT	ORDER_ID
ORDER	SHIPMENT	ORDER_ID
CUSTOMER	REVIEW	CUSTOMER_ID
PRODUCT	REVIEW	PRODUCT_ID

7. Summary

The ER Diagram successfully represents all nine entities, their attributes, relationships, primary keys, foreign keys, and cardinalities required for the **ShopNest – E-Commerce Order Management Database System**. The conceptual design accurately reflects the business requirements and provides a clear understanding of data organization.

The ER Diagram minimizes redundancy, maintains referential integrity, and serves as the basis for relational schema design and SQL database implementation.

8. Conclusion

The **Entity Relationship Diagram** provides a complete conceptual representation of the **ShopNest – E-Commerce Order Management Database System**. It clearly illustrates how customers, products, categories, suppliers, orders, payments, shipments, and reviews are interconnected.

This ER design serves as the foundation for creating the relational schema and implementing the database using SQL. It ensures data consistency, integrity, and efficient management of e-commerce operations while supporting future enhancements and business growth.